IGCSE 0607 Practice Questions — E1.5 — Combined Questions & Markscheme

Ordering

Total Questions: 6 — **Total Marks:** 24 Marks

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1 Write the following in order of size, starting with the smallest.

$$\frac{5}{8} \qquad 0.62 \qquad 61.5\% \qquad \frac{3}{5}$$

..... **[2]**

2 **(a)** Insert one of the symbols $<, =$ or $>$ to make each statement true.

(i) $\frac{5}{8} \ldots 0.65$

[1]

(ii) $72\% \ldots \frac{18}{25}$

[1]

(b) Write the following in order of size, starting with the smallest.

$$0.55 \qquad \frac{6}{11} \qquad 54\% \qquad \frac{5}{9}$$

..... **[2]**

3 **(a)** Given that $0 < x < 1$, write the following expressions in order of size, starting with the smallest.

$$x \qquad x^2 \qquad \frac{1}{x} \qquad \sqrt{x}$$

..... **[2]**

(b) Given that $x < -1$, write the following expressions in order of size, starting with the largest.

$$x \qquad x^3 \qquad x^{-2} \qquad \frac{1}{x}$$

..... **[2]**

4 **(a)** Given that $x = -0.4$, write the following in order of size, starting with the smallest.

$$x \qquad x^2 \qquad x^3 \qquad \frac{1}{x}$$

..... **[2]**

(b) Insert the correct symbol ($<, =$, or $>$) to make the statement true.
Explain your reasoning.

$5\sqrt{3} \ldots 4\sqrt{5}$

[2]

(c) Find the integer m that satisfies the inequality

$$\frac{m}{7} < \frac{5}{9} < \frac{m+1}{7}$$

$m =$ ..... **[3]**

5 Write the following masses in order of size, starting with the largest.

5.2×10^{-4} kg 0.04 g 4.5×10^{3} mg 6×10^{-7} tonnes

..... **[3]**

6 **(a)** Here are four numerical expressions.

$a = 3.1 \times 10^3$
$b = \sqrt{9.8 \times 10^6}$
$c = 5^5$
$d = 56^2$

Work out the value of each expression and write them in order of size, starting with the largest.

..... **[3]**

(b) State the integer n such that

$$n < \sqrt{250} < n + 1$$

$n =$ ..... **[1]**

1 Converts to a common format (e.g. decimals: $0.625, 0.62, 0.615, 0.6$) **[M1]**

$\frac{3}{5}, 61.5\%, 0.62, \frac{5}{8}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 (a) (i) $<$ **[B1]**

(ii) $=$ **[B1]**

(b) M1 for converting at least three values into a comparable form (e.g. percentages: 55%, $54.54\% \ldots$, 54%, $55.55\% \ldots$) **[M1]**

$54\%, \frac{6}{11}, 0.55, \frac{5}{9}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 (a) $x^2, x, \sqrt{x}, \frac{1}{x}$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

(b) $x^{-2}, \frac{1}{x}, x, x^3$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

[Back to Question] *[Total Marks: 4]*

4 (a) M1 for correctly calculating at least two values: $x^2 = 0.16$, $x^3 = -0.064$, $\frac{1}{x} = -2.5$ **[M1]**

$\frac{1}{x}, x, x^3, x^2$ **[A1]**

(b) M1 for squaring both terms to get 75 and 80, OR expressing as $\sqrt{75}$ and $\sqrt{80}$ **[M1]**

$<$ **[A1]**

(c) M1 for identifying a common denominator, e.g. cross multiplying to get $9m < 35$ **[M1]**

M1 for $35 < 9(m+1)$ leading to $26 < 9m$ **[M1]**

3 **[A1]**

[Back to Question] *[Total Marks: 7]*

5 Converts all to a common unit, e.g. grams: 0.52 g, 0.04 g, 4.5 g, 0.6 g **[M2]**

(M1 for 2 correct conversions to a common unit)

4.5×10^3 mg, 6×10^{-7} tonnes, 5.2×10^{-4} kg, 0.04 g **[A1]**

[Back to Question] *[Total Marks: 3]*

6 (a) B1 for at least two correct evaluated values ($a = 3100$, $b = 3130.49 \ldots$, $c = 3125$, $d = 3136$) **[B1]**

B1 for all four evaluated correctly **[B1]**

d, b, c, a **[A1]**

(b) 15 **[B1]**

[Back to Question] *[Total Marks: 4]*